

2.04×	+31 pts CAR	−8 dB SNR	cosθ ≈0.75	SBE=1.00	0 contradictions
Semantic throughput vs Shannon (S1)	Over DeepSC at −6 dB SNR (S2)	URLLC vs classical 5G NR (S5)	Semantic security boundary (S6)	Shannon recovered at ADI=0 (S1)	Across all 12 simulations

Sim	Simulation Name	Falsifiable Claim	Primary Metric	Key Result Confirmed	6G / IMT-2030 Relevance	Status
S1	Semantic Capacity Bound	Cs >= C_Shannon under shared context	SBE = Cs/C_Shannon SSNR, ESA	SBE=1.00 at ADI=0 — Shannon recovered exactly Peak SBE=2.04x at ADI=1, SNR=20 dB	ADI as new 6G KPI alongside BER and latency	DONE
S2	DeepSC Benchmark	Anchor decoding beats DeepSC at all SNR	CAR, BLEU, CFI, IPI, BUE	+31 pts CAR over DeepSC at -6 dB SNR +14% BUE at 20 dB. BLEU converges above 12 dB	Semantic coverage beats syntactic at resource limits	DONE
S3	Anchor Dynamics & Super-Recovery	Gamma(t) dynamic; resolved misalignment super-recovers Cs	ARS(t), ARC Cs(t), AER	ARC=1.17. Post-resolution Cs > pre-misalignment Cs No classical analogue exists	Cs(t) as dynamic channel state in 6G	DONE
S4	Semantic Field Geometry	Cauchy-Riemann conditions empirically computable	CRCS, CIDI SRM	CRCS=0.91 (high anchor) vs 0.38 (low anchor) SRM=5.62 — persistent singularities at low anchor	Singularity maps enable semantic scheduler in 6G	DONE
S5	6G V2X URLLC	LPG achieves URLLC at lower SNR than 5G NR	CAR vs SNR Rayleigh+Doppler	CAR>0.95 at -8 dB below 5G NR Half transmit power. -4 dB below DL-SemCom	Addresses IMT-2030 URLLC <0.1 ms latency	DONE
S6	Cognitive Security (IBC)	Intent encryption produces sharp security boundary	Decryption rate vs cos(theta)	<3% decryption at cos(theta)<0.4 >97% at cos(theta)>0.9. Boundary at cos(theta)~0.75	New 6G security primitive complementary to AES-256	DONE
S7	Multi-Agent Consensus	LPG routing achieves near-constant consensus time as N grows	T_c, GAS, SRG scaling beta	beta=0.006 (LPG) vs 0.313 (broadcast) SRG=9.52 at N=32. Consensus 2 rounds vs 19	Semantic scheduling replaces packet routing	DONE
S8	sem Component Robustness	iota most fragile; corruption cascades superadditively	SFI, CFI, CAR cascade ratio	iota threshold k=19.9% vs k=54% for B Cascade ratio 1.09 with iota vs 0.81 without	Identity tags need strongest 6G protection	DONE
S9	Cross-Modal Channel	Semantic capacity is modality-independent (text to image)	CMFI r(Cs, CMFI)	r=0.977 overall. CMFI rises with ADI at SNR=0 dB Anchor compensates for bandwidth constraint	Anchor scheduling for XR/holographic 6G	DONE
S10	Semantic Compression Limit	sem has a minimum viable representation (semantic knee)	SCR, CFI knee iota control	SCR=0.24 vs Shannon ref 0.50 — 2.04x compression iota incompressible. CAR collapses after knee	74% payload cut for IoT 6.3x device density	DONE
S11	Adaptive Feedback (SAE)	4-bit feedback drives alignment >0.90 where ARQ fails	FEG, cos(theta) convergence rate	FEG=1.52 at 4-bit feedback. 8 rounds to threshold ARQ: 0% convergence. SAE: 100% convergence	Semantic feedback as new 6G uplink primitive	DONE
S12	Framework Boundary Conditions	LPG has a mappable validity envelope	FVR, RMSE semantic inertia	FVR: B1=0.90 (analytic) to B5=0.34 (context collapse) 3 rounds semantic inertia after topic shift	Boundary warnings for channel health monitoring	DONE

WHAT ALL 12 ESTABLISH TOGETHER - Shannon capacity is the ADI=0 limit of Cs — SBE=1.00 confirmed exactly. - Anchor richness substitutes for bandwidth, power, and retransmission together. - Cs(t) is time-varying and super-recoverable — no classical analogue. - The semantic field has a mappable topology; singularities are avoidable.	FRAMEWORK STATUS Architecture + algorithm: Mathematical proofs: Python implementation: Synthetic validation: Real-data validation: Publication target: Complete Complete (converse pending) 1,182 lines — complete 12 simulations · 0 contradictions Next step — designs specified IEEE Trans. Communications	THE CORE THESIS “Semantic communication is the realistic virtual extension of Shannon’s information theory.” Shannon is not replaced — recovered exactly as the degenerate case (ADI=0). Every gain above classical capacity displaces Shannon.
--	--	--